# Microbiology Mid-Term 2025-26 吕镇梅老师班级

**1)** A microbiologist finds a new single-celled organism that lacks a nucleus and membrane-bound organelles. To which domains could it belong?
A) Bacteria only
B) Archaea only
C) Either Bacteria or Archaea
D) Eukarya

**2)** The current three-domain classification of life (Bacteria, Archaea, Eukarya) is based primarily on comparisons of ______.
A) ribosomal RNA gene sequences
B) protein-coding genes for metabolism
C) cell wall composition
D) membrane lipid structure

**3)** All living cells share certain fundamental properties. Which of the following is **NOT** universally present in all living cells?
A) The ability to carry out metabolism
B) The ability to evolve over time
C) The ability to differentiate into multiple cell types
D) The ability to transcribe DNA into RNA

**4)** If you could travel back in time 2.5 billion years, which of the following would you **NOT** expect to find?
A) Photosynthetic bacteria
B) Fossilized stromatolites
C) Multicellular animals
D) Archaea living in extreme environments

**5)** A student needs to isolate bacterial colonies from a mixed sample. Which of the following tools is most appropriate for growing the bacteria on solid medium?
A) Erlenmeyer flask
B) Test tube
C) Petri dish
D) Microscope slide

**6)** Pasteur's swan-necked flask experiment disproved spontaneous generation because:
A) the broth in the flask never came into contact with air

B) boiling the broth killed all microbes, and the curved neck prevented new microbes from entering, so the broth remained sterile
C) the flask was made of a special material that killed microbes
D) the broth was replaced daily

**7)** A patient develops septic shock due to a bacterial infection. The causative agent is most likely a ______.
A) Gram-positive bacterium because lipopolysaccharides (LPS) is abundant in their cell walls
B) Gram-negative bacterium because LPS is present in their outer membrane
C) Archaeon because they produce LPS as an energy source
D) Fungus because their cell walls contain LPS

**8)** Which of the following is the **primary** function of the cytoplasmic membrane in both prokaryotes and eukaryotes?
A) Providing rigid structural support to the cell
B) Serving as a site for protein synthesis
C) Acting as a selective permeability barrier
D) Storing genetic material

**9)** Which of the following structures is **unique** to gram-negative bacteria and is not found in gram-positive bacteria?
A) Peptidoglycan layer
B) Cytoplasmic membrane
C) Periplasmic space
D) Capsule

**10)** Which of the following statements about bacterial endospores is **true**?
A) Sporulation is a method of reproduction that produces two daughter cells.
B) Endospores are easily stained with basic dyes and appear dark inside vegetative cells.
C) Calcium dipicolinate in the endospore core contributes to dehydration and heat resistance.
D) The three stages of sporulation are activation, germination, and outgrowth.

**11)** A virulent strain of *Streptococcus pneumoniae* produces a polysaccharide capsule. This structure helps the bacterium evade host immune cells. However, it does **NOT** provide ______.
A) Resistance to engulfment by macrophages
B) Ability to form biofilms on medical devices
C) Protection against bursting in hypotonic environments
D) Attachment to teeth or other surfaces

**12)** Which of the following statements about bacterial transport systems is **false**?
A) ABC transporters use ATP hydrolysis to move substrates.
B) Group translocation chemically modifies the substrate during transport.
C) Active transport can be driven by the proton motive force.
D) Active transport moves substances down their concentration gradient without energy input.

**13)** Which one is the most fundamental defining feature that distinguishes eukaryotic microbial cells from prokaryotic cells?
A) Containing ribosomes for protein synthesis
B) Possessing a double membrane-enclosed nucleus
C) Cell wall composed of peptidoglycan
D) Using ATP as an energy currency

**14)** Which is the key function of sterols present in eukaryotic cytoplasmic membranes?
A) To catalyze chemical reactions on the membrane surface
B) To store genetic information for cell division
C) To mediate the transport of molecules across the lipid bilayer
D) To provide structural strength and maintain membrane integrity

**15)** Mycelium is defined as ______.
A) an infection structure
B) a sexual fruiting body
C) a mass of hyphae
D) a single fungal spore

**16)** Yeasts are best defined as ______.
A) Multicellular filamentous fungi
B) Unicellular eukaryotic fungi
C) Anaerobic protozoa
D) Photosynthetic algae

**17)** What is the main structural difference between rough endoplasmic reticulum (RER) and smooth endoplasmic reticulum (SER)?
A) RER lacks ribosomes; SER has attached ribosomes
B) RER is continuous with the cell membrane; SER is not
C) RER synthesizes lipids; SER synthesizes proteins
D) RER has attached ribosomes; SER lacks ribosomes

**18)** The endosymbiotic hypothesis states that mitochondria and chloroplasts originated from ______.
A) ancient archaea

B) respiratory and phototrophic bacteria

C) early fungal cells

D) protozoan hosts

**19)** What is the outcome of mitosis?

A) Two genetically identical diploid daughter cells

B) Four genetically distinct haploid gametes

C) One diploid zygote from two haploid cells

D) Conversion of a diploid cell to a haploid cell

**20)** Mitochondria are organelles specialized for which core cellular process?

A) Energy metabolism via respiration and oxidative phosphorylation

B) Protein folding and post-translational modification

C) Lipid synthesis and carbohydrate metabolism

D) Photosynthesis and carbon fixation

**21)** Eukaryotic flagella move via which mechanism?

A) Rotation like a propeller

B) Whiplike motion driven by ATP

C) Sliding via peptidoglycan filaments

D) Amoeboid movement

**22)** Which of the following is not the function of **ethanol** when making a microscopic slide of *Aspergillus flavus*?

A) Dehydrating the fungal sample

B) Washing away impurities on the hyphae

C) Staining the mycelium and spores

D) Assisting in sample fixation

**23)** A bacterium isolated from a deep-sea hydrothermal vent obtains energy by oxidizing hydrogen sulfide ($H_2S$) to elemental sulfur. This type of metabolism is called ______.

A) Chemoorganotrophy

B) Chemolithotrophy

C) Phototrophy

D) Fermentation

**24)** During lactic acid fermentation in *Lactobacillus*, the majority of ATP is synthesized by ______.

A) Oxidative phosphorylation via an electron transport chain

B) Substrate-level phosphorylation

C) Photophosphorylation
D) The citric acid cycle

**25)** A biochemist measures $\Delta G^{0'}$ for a metabolic reaction and finds it to be +250 kJ/mol. To proceed, this reaction must be ______.
A) Coupled to an exergonic reaction to provide energy
B) Allowed to proceed spontaneously
C) Carried out at a higher temperature
D) Catalyzed by a different enzyme

**26)** A bacterium growing aerobically on lactose transfers electrons from lactose to oxygen. In this process, lactose acts as the ______ and oxygen acts as the ______.
A) Oxidizing agent / reducing agent
B) Electron acceptor / electron donor
C) Reducing agent / oxidizing agent
D) Catalyst / product

**27)** The complete oxidation of glucose to $CO_2$ in aerobic respiration involves two major stages: conversion of glucose to pyruvate, followed by oxidation of pyruvate to $CO_2$. These stages are respectively:
A) Glycolysis and the citric acid cycle
B) The citric acid cycle and glycolysis
C) Fermentation and the electron transport chain
D) Glycolysis and fermentation

**28)** A bacterium is able to grow in an anaerobic environment using nitrate as an electron acceptor. This process is called ______.
A) Aerobic respiration
B) Fermentation
C) Anaerobic respiration
D) Photosynthesis

**29)** A researcher monitors a bacterial culture over time. Which measurement directly indicates microbial growth?
A) The average cell length doubles.
B) The optical density (turbidity) of the culture increases.
C) Individual cells become more motile.
D) The culture pH decreases.

**30)** Which of the following best describes the bacterial lag phase?
A) Cells are dying at a constant rate.
B) Cells are actively dividing at maximum rate.
C) Cells are synthesizing necessary enzymes and adapting to the new environment.
D) The number of viable cells is decreasing exponentially.

**31)** A student observes that when she transfers bacteria from a culture that is actively growing in nutrient broth to fresh nutrient broth, the cells start dividing immediately without a pause. This is because:
A) The bacteria are in the death phase.
B) The bacteria are already adapted to the medium and do not need to synthesize new enzymes.
C) The bacteria have formed endospores.
D) The lag phase always occurs regardless of conditions.

**32)** Which of the following microorganisms would **lack** superoxide dismutase?
A) *Escherichia coli* (facultative anaerobe)
B) *Bacillus subtilis* (aerobe)
C) *Clostridium botulinum* (obligate anaerobe)
D) *Streptococcus pyogenes* (aerotolerant anaerobe)

**33)** A researcher wants to isolate a psychrophilic extremophile. Which environment would be most appropriate to sample?
A) A hot spring at 80°C
B) Garden soil at 25°C
C) An Antarctic lake permanently below 0°C
D) Human skin at 37°C

**34)** A clinical microbiologist tests two bacterial strains against the same antibiotic. Strain A has a 25 mm zone, Strain B has a 12 mm zone. Which statement is correct?
A) Strain A is more resistant than Strain B.
B) Strain B is more resistant than Strain A.
C) Both strains are equally susceptible.
D) The zone size does not indicate resistance.

**35)** Which two types of symmetry are most commonly found in virus capsids?
A) Rod-shaped and spherical
B) Helical and icosahedral
C) Bacilliform and cuboidal
D) Complex and enveloped

**36)** During T4 bacteriophage replication, the genome is synthesized as long concatemers. What is the immediate precursor of the mature T4 genome?

A) A circular plasmid-like DNA

B) A unit-length linear DNA produced by host recombination

C) A concatemer that is cleaved at specific sites during packaging

D) An RNA intermediate transcribed from the integrated prophage

**37)** A lysogenized bacterium carries a prophage. Which condition must be maintained to keep the prophage stable without killing the host?

A) The prophage genome replicates independently of the bacterial chromosome.

B) Lytic genes of the prophage are actively transcribed.

C) The prophage integrates into the host genome and replicates passively with it, while lytic genes are repressed.

D) The prophage is excised from the chromosome and replicates rapidly.

**38)** A researcher isolates a new phage that forms clear plaques on a lawn of bacteria. This phage is most likely ______.

A) Temperate

B) Lysogenic

C) Virulent (lytic)

D) Integrated as a prophage

**39)** To determine the number of infectious phage particles in a sample, a microbiologist performs serial dilutions, mixes each dilution with bacteria, and pours the mixture onto an agar plate. After incubation, zones of clearing (plaques) are counted. This method is called:

A) Streak plate

B) Spread plate

C) Plaque assay

D) Broth dilution assay

**40)** A student claims: “Plaque assays can be used to titer both lytic and lysogenic bacteriophages without any additional treatment.” Why is this statement incorrect?

A) Lysogenic phages do not form plaques because they integrate into the host genome.

B) Lytic phages cannot be titered by plaque assays.

C) Plaque assays require a liquid culture, not an agar plate.

D) Lysogenic phages always kill the host instantly, producing too many plaques to count.

**41)** A mutation in the T4 bacteriophage gene encoding T4 lysozyme makes the enzyme non-functional. What would be the most direct consequence of this mutation during infection?

A) The phage tail fibers would not attach to *E. coli*.

B) The phage DNA would be injected but without removing the cell wall, the DNA cannot enter the cytoplasm.
C) The tail sheath cannot contract.
D) The phage cannot produce new lysozyme for host lysis at the end of the infection cycle.

**42)** Which is **NOT** a possible outcome from infection by an animal virus?
A) lysogenic infection
B) persistent infection
C) latent infection
D) transformation

**43)** Which is **NOT** true regarding retroviruses?
A) Reverse transcriptase creates dsDNA from the ssRNA genome.
B) Integrase inserts the viral dsDNA into the host cell's genome as a provirus.
C) Protease excises the provirus from the host cell genome.
D) New nucleocapsids exit the cell by budding.

**44)** Influenza virus has a segmented negative-sense RNA genome. This means that ______.
A) The viral RNA can be translated directly into viral proteins.
B) The virus must carry an RNA-dependent RNA polymerase inside the virion.
C) The viral genome is identical to mRNA.
D) The host cell's DNA-dependent RNA polymerase can transcribe it.

**45)** In bacterial transcription initiation, the specificity of promoter binding is determined by a subunit of RNA polymerase that recognizes consensus sequences at the -10 and -35 regions. This subunit is called ______.
A) Rho factor
B) Sigma factor
C) TATA-binding protein
D) Primase

**46)** A student makes four claims about DNA replication and transcription. Which claim is **false**?
A) DNA serves as the template for both processes.
B) In both processes, the new strand is synthesized 5'→3'.
C) Both processes require a pre-existing 3'-OH primer to initiate.
D) The template strand is antiparallel to the newly synthesized strand.

**47)** A mutation in a bacterial gene alters a stretch of DNA that forms a GC-rich hairpin followed by a poly-U sequence in the RNA. This mutation causes RNA polymerase to read through the normal termination point, producing a longer transcript. This suggests that the mutated region is ______.

A) A ribosome binding site
B) A transcription terminator
C) A promoter
D) An origin of replication

**48)** Two DNA samples have the same length but different base compositions. Sample 1 has 30% G+C, Sample 2 has 60% G+C. Which sample will require a higher temperature to denature (separate strands) completely?
A) Sample 1 (30% G+C)
B) Sample 2 (60% G+C)
C) Both melt at the same temperature because length is equal.
D) Melting temperature is independent of base composition.

**49)** The amino acid leucine is specified by six different codons (UUA, UUG, CUU, CUC, CUA, CUG). This property of the genetic code is known as ______.
A) Wobble
B) Degeneracy
C) Universality
D) Non-overlapping

**50)** A student makes the following statements about DNA replication and transcription. Which statement is correct?
A) Both processes require DNA helicase to unwind the DNA.
B) In transcription, RNA polymerase itself unwinds the DNA locally.
C) Topoisomerase is the main enzyme that separates DNA strands during transcription.
D) DNA gyrase is responsible for unwinding the DNA in both processes.

# Complete Answer Key (1-50)

1. **C**
2. **A**
3. **C**
4. **C**
5. **C**
6. **B**
7. **B**
8. **C**
9. **C**

10. **C**
11. **C**
12. **D**
13. **B**
14. **D**
15. **C**
16. **B**
17. **D**
18. **B**
19. **A**
20. **A**
21. **B**
22. **C**
23. **B**
24. **B**
25. **A**
26. **C**
27. **A**
28. **C**
29. **B**
30. **C**
31. **B**
32. **C**
33. **C**
34. **B**
35. **B**
36. **C**
37. **C**
38. **C**
39. **C**
40. **A**
41. **B**
42. **A**
43. **C**
44. **B**
45. **B**
46. **C**
47. **B**

48. **B**
49. **B**
50. **B**